1 My solutions for exercises

I'm practicing proof writing as well as problem solving here, so the writing style can vary from one solution to another.

1.1 Exercise 2.2.3

1.1.1 (a) $a \geq a$.

Proof. We seek an m such that $a = a + m$. Clearly 0 is such an m, and hence $a \geq a$.

1.1.2 (b) If $a \geq b$ and $b \geq c$, then $a \geq c$.

Proof. Let m and n be natural numbers such that $a = b + m$ and $b = c + n$. Then we have $a = (c + n) + m$. By the associative law for addition, $a = c + (n + m)$. Therefore $a \geq c$.

1.1.3 (c) If $a \geq b$ and $b \geq a$, then $a = b$.

Proof. Let m and n be natural numbers such that $a = b + m$ and $b = a + n$. Then we have $a = (a+n)+m$ and $a = a+(n+m)$, and so $n+m$ must be zero. Thus $n = m = 0$ by Corollary 2.2.9. Substituting 0 for m into $a = b+m$, we conclude that $a = b$.

1. Lemma. If $a + m = a$ then $m = 0$.

   Proof. $a + m = a = a + 0$. Using the cancellation law gives $m = 0$.

1.1.4 (d) $a \geq b$ iff $a + c \geq b + c$.

Proof.

1. (i) If $a \geq b$ then $a + c \geq b + c$.

   Let $a = b + m$ for some natural number m. Then $a + c = (b + c) + m$, hence $a + c \geq b + c$.

2. (ii) If $a + c \geq b + c$ then $a \geq b$.

   Since $a + c \geq b + c$, there exists a natural number m such that $a + c = (b+c)+m$. Then $a+c = (b+m)+c$. By the cancellation law $a = b+m$. Hence $a \geq b$.

1.1.5 (e) $a < b$ iff $a{+}{+} \leq b$.

Proof. First, suppose $a < b$. Then there exists a natural number m such that

$$b = a + m$$

Since $a < b$ implies $a \neq b$, $m \neq 0$. Thus by Lemma 2.2.10, there is a natural number n such that $n{+}{+} = m$. Hence

$$b = a + (n{+}{+}) = (a + n){+}{+} = (a{+}{+}) + n$$

Therefore $a{+}{+} \leq b$.

Conversely, suppose that $a{+}{+} \leq b$. Let m be a natural number such that $b = (a{+}{+}) + m$. Then

$$b = (a{+}{+}) + m = (a + m){+}{+} = a + (m{+}{+})$$

Suppose, for contradiction, that $a = b$. Then

$$a = a + (m{+}{+})$$

and cancelling a gives $m{+}{+} = 0$. But that contradicts Axiom 2.3, which says $m{+}{+} \neq 0$. Thus $a \neq b$.

Since $b = a + (m{+}{+})$, we conclude that $a < b$.

1.1.6 (f) $a < b$ iff $b = a + d$ for some positive number d.

Proof. First, assume that $a < b$. Then $b = a + d$ for some natural number d. Suppose, for contradiction, that $d = 0$. then $b = a + 0 = a$, contradicting the fact that $a \neq b$. Hence $d \neq 0$, and we conclude d is positive.

Conversely, assume that $b = a + d$ for some positive number d. If $a = b$, $d = 0$ by the same argument used in (e), contradicting the fact that $d \neq 0$. Thus $a \neq b$ and we conclude that $a < b$.

1.2 Exercise 2.2.4

Proposition 2.2.13. Let a and b be natural numbers. Then exactly one of the following statements is true: $a < b$, $a = b$, $a > b$.

Proof. We first prove that at most one of the statements is true. If $a = b$, by definition $a < b$ and $a > b$ is false. If $a < b$ or $a > b$, then by definition $a \neq b$. Finally, if $a < b$ and $a > b$, then $a = b$ by Proposition 2.2.12.(c), a contradiction. Hence at most one of the statements is true.

We now prove that at least one of the statements is true. We proceed by induction on a. Consider the base case $a = 0$. Since $b = 0 + b$, we have $0 \leq b$ for all b. Thus either $0 < b$ or $0 = b$, and the base case is done. Now suppose inductively that $a < b \vee a = b \vee a > b$. We need to show that the proposition still holds on $a{+}{+}$. If $a = b$, we have $a{+}{+} = b{+}{+} = b + 1$. Since $a{+}{+} \neq b$, $a{+}{+} > b$. If $a < b$, $a{+}{+} \leq b$ by Proposition 2.2.12.(e). Then either $a{+}{+} < b$ or $a{+}{+} = b$, and in either case we are done. Finally, if $a > b$ then for some natural number m, $a = b + m$. Now $a{+}{+} = (b+m){+}{+} = b + (m{+}{+})$. Hence $a{+}{+} > b$. This closes the induction.

1.3 Exercise 2.2.5.

Proposition 2.2.14 (Strong principle of induction). Let m_0 be a natural number, and let $P(m)$ be a property pertaining to an arbitrary natural number m. Suppose that for each $m \geq m_0$, we have the following implication: If $P(m')$ is true for all natural numbers $m_0 \leq m' < m$, then $P(m)$ is also true. (In particular, this means that $P(m_0)$ is true, since in this case the hypothesis is vacuous.) Then $P(m)$ is true for all natural numbers $m \geq m_0$.

The solution became unnecessarily long here because I tried to give a different proof for each lemma.

1.3.1 Lemma a. $a \neq 0 \to \exists m(N(m) \wedge a = 1 + m)$.

Proof. Since a is positive, by Lemma 2.2.10 there exists a natural number m such that $a = m{+}{+}$. Then we have $a = m{+}{+} = 0 + m{+}{+} = (0 + m){+}{+} = (0{+}{+}) + m = 1 + m$.

1.3.2 Lemma b. $m \neq 0 \wedge n \neq 0 \to m + n \neq 1$.

Proof 1. Suppose $m + n = 1$. by Lemma 2.2.10, there exist natural numbers p and q such that $p{+}{+} = m$ and $q{+}{+} = n$. then $m + n = (p{+}{+}) + (q{+}{+}) = ((p + q){+}{+}){+}{+} = 1 = 0{+}{+}$. by Axiom 2.4. we infer $(p + q){+}{+} = 0$, contradicting axiom 2.3., i.e., $(p + q) \neq 0$. hence $m + n \neq 1$.

Proof 2. Suppose $m + n = 1$. by Lemma 2.2.10, there exists a natural number p such that $p{+}{+} = m$. then $m + n = (p{+}{+}) + n = (p + n){+}{+} = 1 = 0{+}{+}$. by axiom 2.4 we infer $p + n = 0$. but since n is positive, $p + n \neq 0$ By Proposition 2.2.8, a contradiction. Hence $m + n \neq 1$.

1.3.3 Lemma c. $a < b{+}{+} \wedge a \neq b \to a < b$.

Proof 1 (direct). By Proposition 2.2.12.(f), $b{+}{+} = a + n$ for some positive number n. Since n is positive, by Lemma a there exists an m such that $n = m + 1$. Now $b{+}{+} = b + 1 = a + n = a + (m + 1) = (a + m) + 1$. Cancelling 1 from both sides yields $b = a + m$. Thus $b \geq a$. Since we also have $a \neq b$, we conclude that $b > a$.

Proof 2 (indirect). Suppose $a < b$ is false. Then by Proposition 2.2.13, $a > b$ must hold. Now choose two positive numbers n and m such that $b{+}{+} = a+n$ and $a = b+m$. Now $b{+}{+} = b+1 = a+n = (b+m)+n = b+(m+n)$. Cancelling b from both sides gives $m + n = 1$. But this contradicts Lemma b, since m and n are both positive. Hence $b > a$.

1.3.4 Proof

Proof. Let $Q(n)$ be the property that $P(m)$ is true for all $m_0 \leq m < n$. We prove by induction that $Q(n)$ is true for every natural number n.

If $n = 0$ then $Q(0)$ is vacuously true, since there's no m such that $m < 0$.

Now suppose inductively that $Q(n)$. Let m satisfy $m_0 \leq m < n{+}{+}$. We must show $P(m)$. We can think of two cases here: $m = n$ and $m \neq n$.

First, consider the case $m = n$. Notice that by the inductive hypothesis, $P(m')$ for all $m_0 \leq m' < n$. It follows that $P(n)$, i.e., $P(m)$, from the hypothesis of the theorem, and we are done.

Now if $m \neq n$, by Lemma c $m < n$. Thus by the inductive hypothesis we infer that $P(m)$.

This closes the induction, and we have proven that $Q(n)$ holds for every natural number n.

Using this result, we now show the conclusion of the theorem, that is, $P(m)$ for all $m \geq m_0$. For every natural number $m \geq m_0$, $Q(m{+}{+})$ holds, since $m{+}{+}$ is a natural number. Together with the fact that $m_0 \leq m < m{+}{+}$, we conclude that $P(m)$. This demonstrates the strong principle of induction.

1.4 Exercise 2.2.6.

Let n be a natural number, and let $P(m)$ be a property pertaining to the natural numbers such that whenever $P(m++)$ is true, then $P(m)$ is true. Suppose that $P(n)$ is also true. Prove that $P(m)$ is true for all natural numbers $m \leq n$; this is known as the principle of backwards induction.

1.4.1 Initial solution

1. Lemma. If $m \leq 0$ then $m = 0$.

   Proof. Let $m \leq 0$. By definition $0 = m + n$ for some natural number n. But by Corollary 2.2.9, m and n must both be zero. Hence $m = 0$.

2. Solution

   We induct on n. We first prove the principle holds on $n = 0$. Let m be a natural number such that $m \leq 0$. Then m must be 0, and from $P(0)$ we immediately obtain that $P(m)$.

   Now suppose inductively that the principle holds on n. We have to prove the principle on $n{+}{+}$. We first obtain the conclusion of the inductive hypothesis. We have $P(n{+}{+})$, so we infer $P(n)$ from the hypothesis. we have the requirements of backwards induction on n, and thus $P(m)$ is true for every natural number $m \leq n$.

   Now let m be a natural number such that $m \leq n{+}{+}$. If $m = n{+}{+}$, then $P(n{+}{+})$, i.e., $P(m)$ and we are done. Otherwise, $m \leq n$ and from the earlier result we obtain $P(m)$. This closes the induction, and we have demonstrated the principle of backwards induction.

1.4.2 Solution with revised writing & structure

Using the same lemma as above.

Let $Q(n)$ be the property that: "if $P(n)$ is true, then $P(m)$ is true for all $m \leq n$". We prove $Q(n)$ for every natural number n, using induction.

Consider the base case $n = 0$. Let $m \leq 0$. Then m must be 0 by the above lemma, and since $P(0)$, we conclude that $P(m)$.

Suppose inductively that $Q(n)$. We have to prove $Q(n{+}{+})$. From $P(n{+}{+})$ and the definition of P, we have $P(n)$. Thus by the inductive hypothesis, $P(m)$ for all $m \leq n$.

Now let m be a natural number such that $m \leq n{+}{+}$. If $m = n{+}{+}$, then we subtitute m for $n{+}{+}$ in $P(n{+}{+})$, and we are done. Otherwise, $m \leq n$ and thus $P(m)$. This closes the induction.

1.5 The product of two natural numbers is a natural number.

Proof. Let n, m be natural numbers. We induct on n, keeping m fixed.

The base case is obvious, since $0 \cdot m = 0$, and 0 is a natural number.

Now suppose inductively that $n \cdot m$ is a natural number. Then $(n{+}{+}) \cdot m = (n \cdot m) + n$ by Definition 2.3.1. Since $n \cdot m$ and m are natural numbers, $(n \cdot m) + n$ is so by the property of addition. This closes the induction.

1.6 Exercise 2.3.1.

Lemma 2.3.2 (Multiplication is commutative). Let n, m be natural numbers. Then $n \cdot m = m \cdot n$.

1.6.1 Lemma. $n \cdot 0 = 0$.

Proof. We prove by induction. The base case $0 \cdot 0 = 0$ is straightforward from the definition. Now if $n \cdot 0 = 0$, then $(n{+}{+}) \cdot 0 = (n \cdot 0) + 0 = 0 + 0 = 0$.

1.6.2 Lemma. $m \cdot (n{+}{+}) = (m \cdot n) + m$.

Proof. We use induction on m. For the base case, by definition of multiplication we have $0 \cdot (n{+}{+}) = 0 = 0 + 0 = (0 \cdot n) + 0$, and we are done.

Now suppose inductively that $m \cdot (n{+}{+}) = (m \cdot n) + m$. Then by definition of multiplication

$$(m{+}{+}) \cdot (n{+}{+}) = (m \cdot (n{+}{+})) + n{+}{+} = (m \cdot n + m) + n{+}{+}$$

And the properties of addition gives

$$(m \cdot n + m) + n{+}{+} = m \cdot n + (m + n){+}{+} = m \cdot n + n + (m{+}{+})$$

Finally, applying the definition of multiplication we obtain
$(m{+}{+}) \cdot n + m{+}{+}$.
Thus $(m{+}{+}) \cdot (n{+}{+}) = (m{+}{+}) \cdot n + m{+}{+}$. This closes the induction.

1.6.3 Solution

We induct on n. Consider the base case $n = 0$. Then by definition of multiplication and the first lemma, $0 \cdot m = 0 = m \cdot 0$.

Now we prove the inductive step $(n{+}{+}) \cdot m = m \cdot (n{+}{+})$. We start from the inductive hypothesis $n \cdot m = m \cdot n$. We add m to both sides and obtain $(n \cdot m) + m = (m \cdot n) + m$. By definition, LHS is equal to $(n{+}{+}) \cdot m$. Also, the above lemma says that $(m \cdot n) + m$ is equal to $m \cdot (n{+}{+})$. Thus we conclude that $(n{+}{+}) \cdot m = m \cdot (n{+}{+})$, and this closes the induction.

1.7 Exercise 2.3.2

Lemma 2.3.3 (Positive natural numbers have no zero divisors). Let n, m be natural numbers. Then $n \cdot m = 0$ if and only if at least one of n, m is equal to zero. In particular, if n and m are both positive, then nm is also positive.

Proof. First, suppose that $n = 0 \vee m = 0$. If $n = 0$, then by Definition 2.3.1 $n \cdot m = 0 \cdot m = 0$. Otherwise, $m = 0$, and $m \cdot n = 0$ by the same argument. Since multiplication is commutative, that leads to $n \cdot m = 0$.

Conversely, suppose that $nm = 0$, and for contradiction both n and m are positive. Then by Lemma 2.2.10, $n = p{+}{+}$ for some natural p. Now $n \cdot m = (p{+}{+}) \cdot m$, and by definition of multiplication $n \cdot m = p \cdot m + m$. Since $n \cdot m = 0$, we have $p \cdot m + m = 0$. By Corollary 2.2.9 it follows that $m = 0$, a contradiction. Thus at least one of n and m must be zero.

1.8 Proposition 2.3.4

Proposition 2.3.4 (Distributive Law). For any natural numbers a, b, c, we have $a(b + c) = ab + ac$ and $(b + c)a = ba + ca$.

Proof. Since multiplication is commutative, we only have to show the first identity. We induct on a, keeping b and c fixed.

For the base case $a = 0$, we have the following identities:

$0(b + c) = 0 = 0 + 0 = 0b + 0c$

Hence the base case is done.

Now suppose inductively that $a(b + c) = ab + ac$. Then

$$
\begin{aligned}
(a{+}{+})(b + c) &= a(b + c) + b + c && \text{Definition of multiplication} \\
&= (ab + ac) + b + c && \text{Inductive hypothesis} \\
&= (ab + b) + (ac + c) && \text{Commutativity and associativity of addition} \\
&= (a{+}{+})b + (a{+}{+})c && \text{Definition of multiplication}
\end{aligned}
$$

This proves the inductive step, and thus $a(b+c) = ab+ac$ for all natural numbers a, b, c.

1.9 Exercise 2.3.3

Proposition 2.3.5 (Multiplication is associative). For any natural numbers a, b, c, we have $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

Proof. We use induction on a. The base case $a = 0$ is straightforward: both $(0 \cdot b) \cdot c$ and $0 \cdot (b \cdot c)$ are equal to 0 by the definition of multiplication.

Now suppose inductively that $(a \cdot b) \cdot c = a \cdot (b \cdot c)$. We need to show that $((a{+}{+}) \cdot b) \cdot c = (a{+}{+}) \cdot (b \cdot c)$. The left-hand side is $(a \cdot b + b) \cdot c$ by definition, which is by distributive law equal to $(a \cdot b) \cdot c + b \cdot c$. The right-hand side is $a \cdot (b \cdot c) + b \cdot c$. But by the inductive hypothesis $(a \cdot b) \cdot c = a \cdot (b \cdot c)$, so we conclude that both sides are equal. This closes the induction.

1.10 Proposition 2.3.6

Proposition 2.3.6 (Multiplication preserves order). If a, b are natural numbers such that $a < b$, and c is positive, then $ac < bc$.

Proof. By Proposition 2.2.12.(f), there exists a positive number m such that $b = a + m$. Then $bc = (a + m)c = ac + mc$. Since m and c are both positive, $mc > 0$ by Lemma 2.3.3. Thus, using Proposition 2.2.12.(f) again, we can conclude that $bc > ac$.

1.11 Corollary 2.3.7

Corollary 2.3.7 (Cancellation law). Let a, b, c be natural numbers such that $ac = bc$ and c is non-zero. Then $a = b$.

Proof. We proceed by contradiction. Suppose $a \neq b$. Then by the trichatomy of order, either $a > b$ or $a < b$. If $a > b$, by Proposition 2.3.6 $ac > bc$, which contradicts $ac = bc$. We can also obtain the same result when $a < b$. Hence $a = b$.

1.12 Exercise 2.3.4

Prove the identity $(a + b)^2 = a^2 + 2ab + b^2$ for all natural numbers a, b.

Lemma. $a^1 = a$ for all natural number a.

Proof. $a^1 = a^{0++} = a^0 \cdot a = 1 \cdot a = a$.

We now prove the identity $(a + b)^2 = a^2 + 2ab + b^2$ by the following identities:

$$
\begin{aligned}
(a+b)^2 &= (a+b)^1 \cdot (a+b) && \text{Definition of exponentiation}\\
&= (a+b) \cdot (a+b) && \text{Lemma}\\
&= (a+b) \cdot a + (a+b) \cdot b && \text{Distributive law}\\
&= aa + ba + ab + bb && \text{Distributive law again}\\
&= aa + ab + ab + bb && \text{Commutativity of multiplication}\\
&= aa + 1ab + ab + bb && \text{Identity of multiplication}\\
&= aa + 2ab + bb && \text{Definition of multiplication}\\
&= a^1a + 2ab + b^1b && \text{Lemma}\\
&= a^2 + 2ab + b^2 && \text{Definition of exponentiation}
\end{aligned}
$$

.

(Since the exercise comes immediately after the definition of exponentiation, I decided to be verbose here.)

1.13 Exercise 2.3.5

Proposition 2.3.9 (Euclidean algorithm). Let n be a natural number, and let q be a positive number. Then there exist natural numbers m, r such that $0 \le r < q$ and $n = mq + r$.

Proof. We induct on n, keeping q fixed. Consider the base case $n = 0$. Then we can choose $m = 0$ and $r = 0$, since $0 < q$ and $0 = 0q + 0$.

Now suppose inductively that there exist natural numbers m and r satisfying the conditions. We have to find m' and r' such that $0 \le r' < q$ and $n{++} = m'q + r'$.

Since $n = mq + r$, we have $n{++} = (m'q + r){++} = m' + (r{++})$.

By the inductive hypothesis $r < q$, and thus $r{++} \le q$. So we have two cases here: $r{++} < q$ or $r{++} = q$.

If $r{++} < q$, then $m' = m$ and $r' = r{++}$, so we are done. Otherwise $r{++} = q$, and then $mq + (r{++}) = mq + q = (m+1)q$. Hence we can choose $m' = m + 1$ and $r = 0$. This closes the induction.

1.14 Exercise 3.1.1.

Show that the definition of equality in Definition 3.1.4 is reflexive, symmetric, and transitive.

Proof. Let A, B, C be sets. First, for any object x we have $x \in A \leftrightarrow x \in A$, since it's a tautology. Hence $A = A$. If $A = B$ then for any object x, $x \in A \leftrightarrow x \in B$. But that is equivalent to $x \in B \leftrightarrow x \in A$, which leads to $B = A$. Finally, suppose that $A = B$ and $B = C$. Then for any object x, $x \in A \leftrightarrow x \in B$ and $x \in B \leftrightarrow x \in C$. Thus we have $x \in A \leftrightarrow x \in C$, which is equivalent to $A = C$.

1.15 Exercise 3.1.2.

Using only Definition 3.1.4, Axiom 3.1, Axiom 3.2, and Axiom 3.3, prove that the sets $\varnothing$, $\{\varnothing\}$, $\{\{\varnothing\}\}$,$\{\varnothing, \{\varnothing\}\}$ are all distinct (i.e., no two of them are equal to each other).

1. $\varnothing$ is not equal to any other set listed. By Axiom 3.3, the other sets contain at least one element: $\varnothing \in \{\varnothing\}$, $\{\varnothing\} \in \{\{\varnothing\}\}$, $\varnothing \in \{\varnothing, \{\varnothing\}\}$. But by Axiom 3.2, $\varnothing$ has no elements, specifically $\varnothing \notin \varnothing$ and $\{\varnothing\} \notin \varnothing$. Hence by Definition 3.1.4, no sets listed are equal to $\varnothing$.

2. $\{\varnothing\} \neq \{\{\varnothing\}\}$. By Axiom 3.3, $\{\{\varnothing\}\}$ contains $\{\varnothing\}$, and $\{\varnothing\}$ contains $\{\varnothing\}$ if and only if $\{\varnothing\} = \varnothing$. We already have shown that $\varnothing \neq \{\varnothing\}$. Hence $\{\varnothing\}$ is in $\{\{\varnothing\}\}$ but it is not an element of $\{\varnothing\}$, and by Definition 3.1.4 $\{\varnothing\}$ is not equal to $\{\{\varnothing\}\}$.

3. $\{\varnothing\} \neq \{\varnothing, \{\varnothing\}\}$. By Axiom 3.3, $x \in \{\varnothing, \{\varnothing\}\}$ if and only if $x = \varnothing$ or $x = \{\varnothing\}$. Thus, $\{\varnothing\} \in \{\varnothing, \{\varnothing\}\}$, but we know that $\{\varnothing\}$ is not an element of $\{\varnothing\}$. Thus the two sets are distinct.

4. $\{\{\varnothing\}\} \neq \{\varnothing, \{\varnothing\}\}$. By a similar argument, $\varnothing$ is in $\{\varnothing, \{\varnothing\}\}$, but since $\varnothing \neq \{\varnothing\}$, it's not in $\{\{\varnothing\}\}$. Hence the two sets are also distinct.

1.16 Exercise 3.1.3.

Prove the remaining claims in Lemma 3.1.13.

1.16.1 If a and b are objects, then $\{a, b\} = \{a\} \cup \{b\}$.

We have $x \in \{a, b\} \leftrightarrow x = a \vee x = b$. But $x = a$ is equivalent to $x \in \{a\}$ and $x = b$ is equivalent to $x \in \{b\}$. Thus $x \in \{a, b\} \leftrightarrow x \in \{a\} \vee x \in \{b\}$, and by Axiom 3.4 this is equivalent to $x \in \{a, b\} \leftrightarrow x \in \{a\} \cup \{b\}$. Hence $\{a, b\}$ and $\{a\} \cup \{b\}$ are equal sets.

1.16.2 If A and B are sets, then $A \cup B = B \cup A$.

For any object x we have the following (by Axiom 3.4):

$$\begin{aligned} x \in A \cup B &\leftrightarrow x \in A \vee x \in B \\ &\leftrightarrow x \in B \vee x \in A \\ &\leftrightarrow x \in B \cup A \end{aligned}$$

Hence $A \cup B = B \cup A$ by Definition 3.1.4.

1.16.3 If A is a set, then $A \cup A = A \cup \varnothing = \varnothing \cup A = A$.

Let x be an object. Then $x \in A \cup A$ if and only if $x \in A$ or $x \in A$, which is equivalent to $x \in A$. Thus $A \cup A$ and A are equal sets.

Also, $x \in A$ is equivalent to $x \in A \vee x \in \varnothing$, since $x \in \varnothing$ is always false by Axiom 3.2. So A and $A \cup \varnothing$ are equal.

Finally, by the previous argument we know that the union operation for sets is commutative, so $A \cup \varnothing$ and $\varnothing \cup A$ are also equal.

1.17 Exercise 3.1.11.

Show that the axiom of replacement implies the axiom of specification.

Proof. Let A be a set, and let $P(x)$ be a property pertaining to the elements of A. We have to show that there exists a set X such that for any object x, $x \in X$ if and only if $x \in A$ and $P(x)$.

Let $Q(x, y)$ be the statement "$x = y$ and $P(x)$".

We claim that for each $x \in A$, there is at most one y for which $Q(x, y)$ is true. Let $x \in A$, and let y_1, y_2 satisfy $Q(x, y_1)$ and $Q(x, y_2)$. Then $x = y_1$ and $x = y_2$, which then implies $y_1 = y_2$. Hence our claim is true, and by Axiom of replacement there exists a set Y such that for any object z, $z \in Y$ if and only if $Q(x, z)$ for some $x \in A$.

We now prove that Y is the X we are seeking, i.e., for every z, $z \in Y$ if and only if $z \in A$ and $P(z)$.

First, suppose $z \in A$ and $P(z)$. To show that $z \in Y$, we have to find an $x \in A$ such that $Q(x, z)$ holds. Indeed, $Q(z, z)$ holds, since $z = z$ and $P(z)$. Thus, z is such an x, and therefore $z \in Y$.

Conversely, let $z \in Y$. Then there exists an $x \in A$ such that $Q(x, z)$, i.e., $x = z$ and $P(x)$. Since $x = z$, it follows that $P(z)$ and $z \in A$.

Hence, Y is the set we were looking for, and this completes our proof.